METRIC WALL CROSSING IN RANK-THREE K3 BUNDLE MODULI

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Abstract. We determine the local harmonic L^2 metric near a stability wall in a complex twelve-dimensional moduli space of rank-three bundles on a projective K3 surface. The central bundle is the sum of two copies of one spherical line bundle and a second spherical line bundle, with five-dimensional cross-extension spaces; the central Kähler metric is Ricci-flat. We compare the actual chamber metrics with the standard hyperkähler quotient metrics on $T^*\operatorname{Gr}(2,5)$ and $T^*\operatorname{Gr}(3,5)$, with relative error quadratic in the deformation size. The comparison is uniform through changes of stabilizer and permits unrelated rates of approach to the wall and its singular strata. It gives the exceptional-fibre collapse scale, complete pointed Gromov–Hausdorff limits after rescaling, quantitative matching to the wall metric, and intrinsic distance asymptotics to the contracted loci. The proof corrects the real moment equation within complex orbits with a bound that remains uniform at the reducible strata.

1. Introduction

Variation of stability changes the birational geometry of moduli spaces of sheaves on a K3 surface. Near a polystable sheaf, an Ext-quiver can describe the analytic contraction and its exceptional strata. On the locally free locus, the Hermitian–Einstein correspondence gives another geometric structure: a tangent vector has the L^2 norm of its harmonic Kodaira–Spencer representative. An analytic quiver description does not identify this metric. In particular, it does not determine its behavior when several exceptional strata approach one another at different rates.

We establish such a metric comparison for a rank-three wall crossing whose local complex geometry is the stratified Grassmannian Mukai flop. The wall contains an eight-dimensional stratum with projective-plane fibres and a more reducible point with a Grassmannian fibre. The comparison holds on a full neighborhood containing both strata. It gives relative asymptotics for intrinsic distances, the complete metric obtained by rescaling the contracting region, and a quantitative comparison with the surrounding wall metric.

1.1. **The moduli problem and metric convention.** Section 2 constructs a smooth genus-four K3 surface X containing smooth rational curves D_1, D_2 such that

$$D_1^2 = D_2^2 = -2, \qquad D_1 D_2 = 5, \qquad H = D_1 + D_2 \text{ is ample.} \tag{1}$$

Put $L_i = \mathcal{O}_X(D_i)$ and

$$E_0 = (L_1 \otimes \mathbb{C}^2) \oplus L_2, \qquad v = (3, 2D_1 + D_2, 0), \qquad D = D_1 - D_2.$$

We fix the holomorphic determinant $L_1^2 L_2$. The Mukai vector v is primitive, $v^2 = 10$, and $c_2(E_0) = 8$. At the polarization H, the bundle E_0 is both slope-polystable and Gieseker-polystable.

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Choose a smooth path of Kähler forms ω_t in $H + tD$, for $|t|$ small, with ω_0 Ricci-flat, whose existence in H follows from Yau's theorem [Yau78]. Let $\mathcal{M}_t^{\mathrm{bun}}$ be the moduli space of slope-polystable locally free bundles of vector v and the specified determinant. We use its Hermitian–Yang–Mills connection topology. On each smooth orbit-type stratum, where the conjugacy class of the effective compact stabilizer is fixed, its metric g_t is the harmonic Dolbeault metric, with

$$\|a\|^2_{\omega_t,h} = \int_X \mathrm{tr}(ia^\dagger \wedge a) \wedge \omega_t. \tag{2}$$

There is no division by volume. The real connection variation $a - a^\dagger$ has squared norm twice (2). The determinant line has its ω_t-Hermitian–Einstein metric; changing a scalar bundle metric does not affect the endomorphism norms in (2).

The length distance d_t is defined using continuous paths that are piecewise smooth on the smooth orbit-type strata, with local gauge lifts. Finite-length limiting endpoints at lower strata are allowed. Paths used to define d_t may leave any prescribed coordinate neighborhood; they remain in the locally free moduli component. We do not impose this metric on non-locally-free sheaves.

1.2. **The comparison theorem.** We first specify the finite-dimensional quotient metric appearing in the result. Let

$$\mathcal{V} = \mathrm{Hom}(\mathbb{C}^2, \mathbb{C}^5) \oplus \mathrm{Hom}(\mathbb{C}^5, \mathbb{C}^2), \qquad x = (A, B),$$

with Euclidean Hermitian norm $\|x\|^2 = \mathrm{tr}\, A^*A + \mathrm{tr}\, BB^*$. The group $G = GL_2(\mathbb{C})$ acts by $(A, B) \mapsto (Ag^{-1}, gB)$, with compact form $K = U(2)$. Define

$$M_\tau = \{BA = 0,\ A^*A - BB^* = \tau I_2\}/K, \qquad \tau = \tau(t) = -\frac{28\pi}{3}t. \tag{3}$$

Write g^{q}_τ and d^{q}_τ for its flat quotient metric and length distance, and $r_\tau([A, B]) = (\|A\|^2 + \|B\|^2)^{1/2}$. At $\tau = 0$ these are understood on the stratified quotient.

The invariant $N = AB$ takes values in

$$\mathcal{N} = \{N \in \mathrm{End}(\mathbb{C}^5) : N^2 = 0,\ \mathrm{rank}\, N \le 2\}. \tag{4}$$

For $\tau > 0$, $M_\tau = T^* \mathrm{Gr}(2, 5)$; for $\tau < 0$, $M_\tau = T^* \mathrm{Gr}(3, 5)$; and $M_0 = \mathcal{N}$. The quotient maps $\pi_\tau : M_\tau \to \mathcal{N}$ are the two incidence resolutions. Their fibres are points over rank two, $\mathbb{P}^2$ over rank one, and the full Grassmannians over zero. Section 3 recalls these identifications with their metric normalizations.

Theorem 1.1. *There are $R, t_0, C > 0$ such that, for $|t| < t_0$, there are homeomorphisms*

$$f_t : \{r_{\tau(t)} < R\} \subset M_{\tau(t)} \longrightarrow U_t \subset \mathcal{M}_t^{\mathrm{bun}}$$

onto open subsets. They are smooth and biholomorphic on the corresponding smooth complex orbit-type strata. The vertex maps to $[E_0]$ at $t = 0$, and the whole zero section is included for all sufficiently small nonzero t. Every point of U_t is slope-stable when $t \neq 0$.

On every smooth piece and for every tangent vector,

$$(1 - C(|t| + r_\tau^2))g^{\mathrm{q}}_\tau \ \le\ f_t^* g_t \ \le\ (1 + C(|t| + r_\tau^2))g^{\mathrm{q}}_\tau. \tag{5}$$

The constants are independent of the positive singular values of AB and of their ratios to each other and to $|t|$.

For some $c, C' > 0$, if $r_\tau(p), r_\tau(q) \le \rho \le cR$, then

$$(1 - C'(|t| + \rho^2))d^{\mathrm{q}}_\tau(p, q) \le d_t(f_t(p), f_t(q)) \le (1 + C'(|t| + \rho^2))d^{\mathrm{q}}_\tau(p, q). \tag{6}$$

The middle distance allows paths throughout the locally free moduli component.

The parameter in (3) follows from the Chern degrees and the harmonic pairing; it is not a choice of scaling exponent. Since $r_\tau^2 \geq 2|\tau|$, the error in (5) is also $O(r_\tau^2)$. At the wall, the rank-one piece parametrizes bundles $L_1 \oplus F$, where F is stable of Mukai vector $(2, H, 0)$. It meets the more reducible bundle E_0. Thus the uniformity includes a genuine change from effective circle symmetry to effective $U(2)$ symmetry.

For the integral ample classes $mH \pm D$, with m sufficiently large, $c_1(E_0)(mH \pm D) = 9m \mp 7$ is coprime to three. Slope and Gieseker stability therefore coincide for these invariants. The two chamber neighborhoods occur in the locally free loci of the ordinary projective sheaf moduli spaces. The contraction diagram asserted here is local and analytic. It does not require a global morphism to the ordinary H-Gieseker wall moduli space.

1.3. **Distances and rescaled geometry.** Let $s_1 \geq s_2 \geq 0$ be the singular values of $N = AB$. For $\kappa > 0$ set

$$b_i^2 = \frac{\sqrt{\kappa^2 + 4s_i^2} - \kappa}{2}, \qquad F_\kappa(b) = \int_0^b \sqrt{\frac{\kappa + 2u^2}{\kappa + u^2}}\, du. \tag{7}$$

For $\kappa = 0$ put $b_i = \sqrt{s_i}$ and $F_0(b) = \sqrt{2}\, b$. Take a slightly larger chart than the one in Theorem 1.1, so that closed radius subsets are defined. Let

$$\mathcal{E}_t = f_t(\pi_\tau^{-1}(0)), \qquad \mathcal{D}_t = f_t(\pi_\tau^{-1}\{\operatorname{rank} N \leq 1\} \cap \{r_\tau \leq R\}).$$

These are the deepest fibre and the local contracted locus; at $t = 0$ they become the vertex and the closure of the reducible stratum.

Theorem 1.2. *For x of radius $r = r_\tau(x) \leq cR$, with $\kappa = |\tau|$,*

$$d_t(f_t(x), \mathcal{E}_t) = (1 + O(|t| + r^2))\sqrt{F_\kappa(b_1)^2 + F_\kappa(b_2)^2}, \tag{8}$$

$$d_t(f_t(x), \mathcal{D}_t) = (1 + O(|t| + r^2))F_\kappa(b_2). \tag{9}$$

The errors are multiplicative and uniform in all the indicated scales. At zero distance the formulas mean equality to zero.

The metric induced on $\mathcal{E}_t$ is $|\tau|$ times the standard Grassmannian quotient metric, with relative error $O(|t|)$. On the $\mathbb{P}^2$ fibre over a rank-one matrix of singular value $s > 0$, the induced metric is $|\tau|$ times the Hopf-quotient Fubini–Study metric, with relative error $O(|t| + s)$.

In particular,

$$d_0(f_0(N), \mathcal{D}_0) = (1 + O(r_0^2))\sqrt{2s_2}.$$

For nonzero τ, the exact model profile crosses from $s_2/\sqrt{|\tau|}$ when $s_2 \ll |\tau|$ to $\sqrt{2s_2}$ when $|\tau| \ll s_2$. Formula (9) controls the intermediate ratios and remains relative even if this distance is arbitrarily small compared with r. Fibre metrics in the theorem are induced metrics; their intrinsic distances are not identified with unrestricted ambient distances between fibre points.

Corollary 1.3. *Fix a zero-section point in M_{+1} or M_{-1}. On the corresponding side of the wall, dividing g_t by $\kappa = |\tau(t)|$ gives the complete pointed length-space limit $(M_{+1}, g_{+1}^{\mathrm{q}})$ or $(M_{-1}, g_{-1}^{\mathrm{q}})$. More precisely, the maps $\iota_t(\widehat{x}) = f_t(\sqrt{\kappa}\,\widehat{x})$ satisfy*

$$(1 - C\kappa(1 + \widehat{r}^2))g_{\pm 1}^{\mathrm{q}} \leq \kappa^{-1}\iota_t^* g_t \leq (1 + C\kappa(1 + \widehat{r}^2))g_{\pm 1}^{\mathrm{q}}. \tag{10}$$

This holds on expanding domains $\widehat{r} \leq R_t$ whenever $R_t \to \infty$ and $\kappa R_t^2 \to 0$, and implies pointed Gromov–Hausdorff convergence on every bounded pointed ball.

On the common rank-two locus, using N to identify the chamber with the wall, the actual chamber and wall metrics have relative difference

$$O\left(|t| + r_0^2 + \frac{|\tau|}{s_2}\right) \tag{11}$$

when $|\tau|/s_2$ is sufficiently small.

The abstract Riemannian spaces M_{+1} and M_{-1} are isometric by adjoint interchange. Their contractions over the fixed matrix N are different. The complete metrics in the corollary are classical; the assertion identifies them as limits of the actual harmonic metric. Neither the corollary nor the tensor estimate asserts smooth convergence or curvature bounds.

1.4. **Relation to previous work and the analytic argument.** The complex geometry uses established K3 deformation theory. Budur–Zhang [BZ19, Theorem 1.2] prove formality for a sheaf polystable with respect to any ample polarization. Arbarello–Saccà [AS18, Propositions 4.4 and 6.1] give the equivariant analytic quadratic germ and its quiver interpretation. Their adjacent-polarization wall-crossing theorem concerns pure one-dimensional sheaves; it is not used for the rank-three bundles here. Buchdahl–Schumacher [BS21, BS22] construct harmonic analytic deformation quotients. Delloque's simultaneous deformation theory [Del25, Propositions 4.3–4.6 and Theorem 7.4] supplies corrected HYM equations and an open actual moduli neighborhood on the integrable locus, without an abelian-stabilizer assumption.

Local metric comparison in reduction is also not new. Sjamaar [Sja95, arXiv version, Lemma 2.13 and its proof] compares the reduced Kähler metric with its flat holomorphic slice metric on the top stratum near a chosen quotient point. Sjamaar–Lerman's singular-reduction theory [SL91], using the classical symplectic normal form, and Mayrand's complex Hamiltonian normal form [May22, Theorems 1.3 and 1.4] preserve structures different from the full Riemannian metric. A real symplectic coordinate change need not preserve an independently imposed complex integrability equation. Namikawa–Odaka [NO26, Remark 6.14] discuss the stronger metric comparison problem in the setting of symplectic singularities; their isolated-singularity and finite hyperkähler quotient results do not directly give the simultaneous estimate in Theorem 1.1.

The complete cotangent-Grassmannian metric and its quotient description are classical [BG96]. Qualitative Gromov–Hausdorff continuity through suitable projective Calabi–Yau flops is proved by Rong–Zhang [RZ11, Theorem 1.2]. Actual Weil–Petersson forms on bundle moduli and their extensions are studied in [BS21, BS23]; explicit metrics and degenerations for framed bundles on the blown-up plane appear in [BS26]. Gauge slices and reduction by the residual finite-dimensional group are part of the earlier Yang–Mills moduli theory [GP89]. We do not claim these constructions or the finite Grassmannian scaling as new.

The additional issue is to compare the *actual* real moment levels on the exact obstructed complex locus, with an error independent of the smallest positive infinitesimal gauge direction. A usual implicit-function bound can lose that direction. We replace it by a projection estimate. If $\mu = A^*A - BB^*$ and ρ is the centered corrected moment map, then for every orthogonal projection P on $\mathbb{C}^2$,

$$|\operatorname{tr}(P(\rho - \mu))| \le \epsilon\big(\|AP\|^2 + \|PB\|^2\big).$$

The right side is the squared norm of the circle action generated by P. Testing this estimate on spectral projections of the positive Hermitian factor of a complex gauge transformation bounds that factor uniformly through rank loss. Projection-relative curvature estimates and

finite-dimensional balancing by spectral projections were developed for repeated stable factors in [Wue26a, Wue26c]. Here we adapt this method to the exact obstructed K3 deformation locus and compare different real moment levels. Section 5 gives the required argument in full.

The harmonic norm requires a separate identification. We first minimize over the uniformly coercive complement of the parallel gauge algebra, as in the complementary-gauge method of [Wue26b]. The result is a smooth positive Hermitian form on the full deformation space. A central automorphism acting by minus the identity removes its linear term. Only then do we minimize over the remaining complex orbit directions. Derivatives of the real balancing transformation are exact gauge terms and disappear from that infimum. Finally, a radius estimate excludes shorter paths leaving the chart. These steps give the actual intrinsic conclusions, rather than merely a comparison of norms on an assumed common locus.

The result is local and concerns locally free bundles. It gives no global completion theorem, no bubbling description, and no general metric normal-form theorem for hyperkähler singularities.

The polynomial calculations of Section 2 and the distance profile in Figure 1 can be reproduced from the source accompanying this paper at `https://github.com/bwuebben/stability-degeneration`.

2. A K3 surface and its stability wall

2.1. **An explicit surface.** In $\mathbb{P}^4$ with coordinates $[x_0 : \cdots : x_4]$, put

$$\begin{aligned} q &= x_0x_3 - x_1x_2 + x_4^2, \\ c_0 &= x_0^3 + x_0x_1^2 + 2x_0x_2^2 + 5x_1x_2x_3 + 4x_0x_1x_2 + 3x_1^2x_3 + 2x_2^2x_3, \\ c &= c_0 + x_4(x_0^2 + x_3^2 + x_4^2), \qquad X = V(q, c). \end{aligned} \tag{12}$$

Proposition 2.1. *The surface X is a smooth K3 surface. Its hyperplane section $x_4 = 0$ is the union of two smooth rational curves satisfying (1).*

Proof. The quadratic form is nondegenerate, and c is not divisible by q. Smoothness can be checked by the following exact finite computation. Over $\mathbb{F}_{101}$, take the ideal generated by q, c and the ten 2×2 minors of their Jacobian. On each affine chart $x_i = 1$, its reduced Gröbner basis is

i	0	1	2	3	4
basis	[1]	[1]	[1]	[1]	[1].

All the polynomial inputs are printed in (12); the accompanying calculation constructs the minors and uses exact arithmetic in $\mathbb{F}_{101}$. Thus the geometric singular subscheme in that fibre is empty, not merely its set of rational points. The projective Jacobian subscheme over $\operatorname{Spec}\mathbb{Z}$ is proper. If its generic fibre were nonempty, its closed image would contain the generic point and hence every prime, a contradiction at 101. The characteristic-zero Jacobian criterion proves smoothness. Adjunction gives $K_X = \mathcal{O}_X$, and the complete-intersection cohomology sequence gives $H^1(X, \mathcal{O}_X) = 0$. Hence X is K3.

On $q = x_4 = 0$ use Segre coordinates $[x_0 : x_1 : x_2 : x_3] = [au : av : bu : bv]$. The cubic restricts to fg, where

$$f = a(u^2 + v^2) + buv, \qquad g = (a^2 + 2b^2)u + 3abv. \tag{13}$$

The two coefficients in each linear expression have no common zero. Thus $D_1 = V(f)$ and $D_2 = V(g)$ are smooth rational graphs of bidegrees $(1, 2)$ and $(2, 1)$ on $\mathbb{P}^1 \times \mathbb{P}^1$. On $a = u = 1$,

their intersection has

$$-z^4+2z^2+2=0, \qquad b=-\frac{1+z^2}{z}, \qquad z=v.$$

There are four distinct nonzero roots. The fifth intersection has $[a:b]=[u:v]=[0:1]$. In the chart $b=v=1$, substitution $a=-u/(1+u^2)$ into g has leading term $-u$, proving transversality there. These exhaust the intersection number.

At a finite intersection, the normal derivative of c is $x_0^2+x_3^2=1+(1+z^2)^2=4(1+z^2)$ modulo the quartic, hence is nonzero. At the fifth point it is one. The intersections are therefore transverse also as curves on X. Their sum is the hyperplane section, which is ample. Adjunction gives $D_i^2=-2$, and the five crossings give $D_1D_2=5$. In particular $H^2=6$ and $HD_i=3$. □

There is also a noncomputational existence argument for the same configuration. On the smooth quadric threefold $V(q)$, lift a product section of bidegree $(3,3)$ from the hyperplane quadric. Vary its lift by x_4G, where G is quadratic. This linear variation is base-point-free off the hyperplane, so a general member is smooth there by Bertini. Along either component away from its transverse crossings, the tangential derivative is nonzero. At the finitely many crossings, avoid the finitely many affine hyperplanes on which the normal derivative vanishes. The explicit equations above select one smooth member; no assumption on the full Picard group of that member is needed.

2.2. **Bundles, obstruction and polarization.** For the Mukai convention $v(F)=(r,c_1,r+c_1^2/2-c_2)$,

$$v(L_i)=(1,D_i,0), \quad v(E_0)=(3,2D_1+D_2,0), \quad c_2(E_0)=8, \quad v(E_0)^2=10. \tag{14}$$

The class $2D_1+D_2$ pairs to one with D_1, so it is primitive. The stable K3 moduli dimension is $v^2+2=12$ [Muk84]. Nonemptiness in the chambers will follow from the actual local construction, rather than from the dimension formula.

The line bundles have H-degree three and Euler characteristic one. They are stable for every polarization, and their reduced H-Hilbert polynomials agree. The differences $\pm(D_1-D_2)$ are nonzero classes of H-degree zero and cannot be effective. Therefore

$$\operatorname{Hom}(L_1,L_2)=\operatorname{Hom}(L_2,L_1)=0.$$

Serre duality gives vanishing of both cross Ext^2 groups. Since $(D_1-D_2)^2=-14$, Riemann–Roch gives

$$\dim\operatorname{Ext}^1(L_1,L_2)=\dim\operatorname{Ext}^1(L_2,L_1)=5, \qquad \operatorname{Ext}^1(L_i,L_i)=0. \tag{15}$$

Serre-dual bases identify $\operatorname{Ext}^1(E_0,E_0)$ with $\mathcal{V}$. The quadratic Yoneda obstruction has blocks $-BA$ and $\operatorname{tr}(BA)$, up to a common nonzero scalar; its zero locus is $BA=0$. Section 4 establishes both harmonic normalization of these bases and the convergent comparison with the full reduced integrability germ.

The effective automorphism group is $(GL_2\times GL_1)/\mathbb{C}^*\simeq GL_2$. The determinant-one group is a finite cover with an ineffective μ_3 center. A trivial effective stabilizer will mean simplicity of the bundle, not literal absence of determinant-one scalar automorphisms.

For the ample real class $H_t=H+tD$, one has

$$\begin{gathered} H_t^2=6-14t^2, \\ \mu_t(L_1)=3-7t, \qquad \mu_t(L_2)=3+7t, \qquad \mu_t(E_0)=3-\tfrac{7}{3}t, \\ \mu_t(L_1)-\mu_t(E_0)=-\tfrac{14}{3}t, \qquad \mu_t(L_2)-\mu_t(E_0)=\tfrac{28}{3}t. \end{gathered} \tag{16}$$

Thus the slope ordering changes transversely. If m is sufficiently large, $mH \pm D$ is integral ample and $c_1(E_0)(mH \pm D) = 9m \mp 7$ is coprime to three. Equality of slopes with a subsheaf of rank one or two would force that rank to be divisible by three. The standard implications between slope and Gieseker stability then identify all the stable and semistable loci for these rank-three invariants. This arithmetic does not itself give a wall contraction; the local HYM construction below supplies that identification.

3. The Grassmannian quotient and its distances

The complete hyperkähler metric on a cotangent Grassmannian is classical. The conventions below agree with the quotient construction in [BG96, §2.2], where the real parameter is written $2k$. We give the calculations needed for the metric comparison and distance statements, including their behavior at smaller rank.

3.1. **Quotients, radii and strata.** Use $\mathcal{V}, G, K$ from Section 1, and write g^{amb} for its Euclidean real metric. For $h \in \mathrm{Herm}_2$, the compact fundamental vector and its complex rotation are

$$\xi_h = (-Aih, ihB), \qquad I\xi_h = (Ah, -hB). \tag{17}$$

With $\omega^{\mathrm{amb}}(v,w) = g^{\mathrm{amb}}(Iv, w)$,

$$d\,\mathrm{tr}(\mu h)(v) = 2\omega^{\mathrm{amb}}(\xi_h, v), \qquad \mu = A^*A - BB^*.$$

The complex moment equation is $BA = 0$.

Proposition 3.1. *The spaces M_τ have the incidence presentations stated in the introduction. For $\tau \neq 0$ they are smooth complete hyperkähler manifolds. The rank-k stratum of $M_0 = \mathcal{N}$ has complex dimension $2k(5-k)$ for $k = 0, 1, 2$. If $s_1 \geq s_2 \geq 0$ are the singular values of N, then*

$$r_\tau^2 = \sum_{i=1}^{2} \sqrt{\tau^2 + 4s_i^2}. \tag{18}$$

The dilation $x \mapsto ax$ sends τ to $a^2\tau$, N to a^2N, and the quotient metric to $a^2 g_\tau^{\mathrm{q}}$.

Proof. For $\tau > 0$, A is injective. Its image U is a two-plane, and $BA = 0$ gives $\mathrm{im}\, N \subset U \subset \ker N$. Conversely, fix such (U, N) and choose an isometric frame $U_0 : \mathbb{C}^2 \to U$. Write $N = U_0 C$, so $CU_0 = 0$. Set

$$P = \frac{\tau I + \sqrt{\tau^2 I + 4CC^*}}{2}, \qquad A = U_0 P^{1/2}, \qquad B = P^{-1/2} C. \tag{19}$$

Then $BA = 0$, $AB = N$, and $A^*A - BB^* = \tau I$. Conversely, the polar decomposition of any balanced A yields this formula, since $P(P - \tau I) = CC^*$ has the displayed unique positive solution. Changing the frame changes the representative by K. The incidence variety is $T^*\,\mathrm{Gr}(2,5)$. The adjoint interchange $(A, B) \mapsto (B^*, A^*)$ exchanges τ with $-\tau$ and gives the analogous presentation with a three-plane between $\mathrm{im}\, N$ and $\ker N$.

At zero level, $A^*A = BB^* = P$. On its support, polar decomposition gives $A = U_0 P^{1/2}$ and $B = P^{1/2} V_0^*$, with $V_0^* U_0 = 0$. Thus $N = U_0 P V_0^*$ determines its balanced compact orbit. Conversely, a singular-value decomposition of any square-zero matrix of rank at most two has this form and supplies a zero-level representative. The invariant ring is generated by the entries of AB, by the first fundamental theorem for GL_2 [KP96, §2.1–2.2]. Restriction to the reduced invariant subvariety $BA = 0$ preserves these generators, by the Reynolds projection.

Its image is exactly (4), so the reduced affine quotient is $\mathcal{N}$. This argument does not require an unproved normality assertion.

At rank two the incidence planes are $\operatorname{im} N$ and $\ker N$. At rank one they are respectively a line and a two-plane in $\ker N/\operatorname{im} N \simeq \mathbb{C}^3$, giving two dual projective planes. At zero they are the whole Grassmannians. Counting a flag $\operatorname{im} N \subset \ker N$ and the induced isomorphism $\mathbb{C}^5/\ker N \to \operatorname{im} N$ gives dimension $2k(5-k)$.

For nonzero τ the effective compact action is free. The complex equation has surjective differential when A is injective or B surjective, and the real moment has full transverse rank. Thus the quotient is smooth. The level is closed in Euclidean space, and its compact quotient with the induced length metric is complete: a finite-length escaping path would have bounded Euclidean radius and hence an endpoint in a compact part of the level quotient.

The eigenvalues of $P + BB^*$ are $\sqrt{\tau^2 + 4s_i^2}$, proving (18). The dilation assertions follow directly from the quadratic equations and Euclidean metric. □

The two incidence maps over the same N are distinct, although adjoint interchange is an isometry between the abstract Riemannian spaces at opposite parameters. Their complex dimensions are twelve. At zero the radius is $r_0^2 = 2(s_1 + s_2)$.

3.2. **Exact model distances.** Let $E_\tau = \pi_\tau^{-1}(0)$ and $D_\tau = \pi_\tau^{-1}\{\operatorname{rank} N \le 1\}$, including the zero-level meanings. Use b_i, F_κ from (7).

Proposition 3.2. *For the full quotient length spaces,*

$$d_\tau^{\mathrm{q}}(x, E_\tau) = \sqrt{F_\kappa(b_1)^2 + F_\kappa(b_2)^2}, \qquad d_\tau^{\mathrm{q}}(x, D_\tau) = F_\kappa(b_2), \quad \kappa = |\tau|. \tag{20}$$

Both distances have minimizing paths along which r_τ decreases.

Proof. First take $\tau > 0$. Diagonalize BB^* in the polar frame of A. There are orthogonal isometric two-frames U, V with

$$A = U \operatorname{diag}(a_1, a_2), \quad B = \operatorname{diag}(b_1, b_2)V^*, \quad a_i = \sqrt{\tau + b_i^2}.$$

If a b_i vanishes, fill out the corresponding frame in $U^\perp$, which has dimension three. Then $s_i = a_i b_i$, giving the expression for b_i in (7).

The Frobenius singular-value inequality [Mir60] $\sum_i |\sigma_i(X) - \sigma_i(Y)|^2 \le \|X - Y\|^2$ implies that ordered singular values along an absolutely continuous matrix path are absolutely continuous, with their squared speeds bounded by the squared matrix speed. Applying it to A and B yields

$$\|\dot A\|^2 + \|\dot B\|^2 \ge \sum_i (\dot a_i^2 + \dot b_i^2) = \sum_i \left| \frac{d}{du} F_\tau(b_i) \right|^2 \tag{21}$$

almost everywhere. Difference quotients prove this also at collisions; no smooth choice of singular vectors is assumed. Compact gauge changes preserve all the singular values, so horizontal minimization proves that $x \mapsto (F_\tau(b_1), F_\tau(b_2))$ is 1-Lipschitz for quotient length. Its value on E_τ is zero, and its second component vanishes on D_τ. This proves both lower bounds.

For the first upper bound, keep U, V fixed and set $F_\tau(b_i(u)) = uF_\tau(b_i(1))$ for $0 \le u \le 1$. Choose $a_i(u) = \sqrt{\tau + b_i(u)^2}$. The path stays on both moment equations and has constant ambient speed equal to the first expression in (20). It ends on E_τ. For the second upper bound, keep b_1 fixed and vary only b_2 in this way. The radii decrease. Their real diagonal

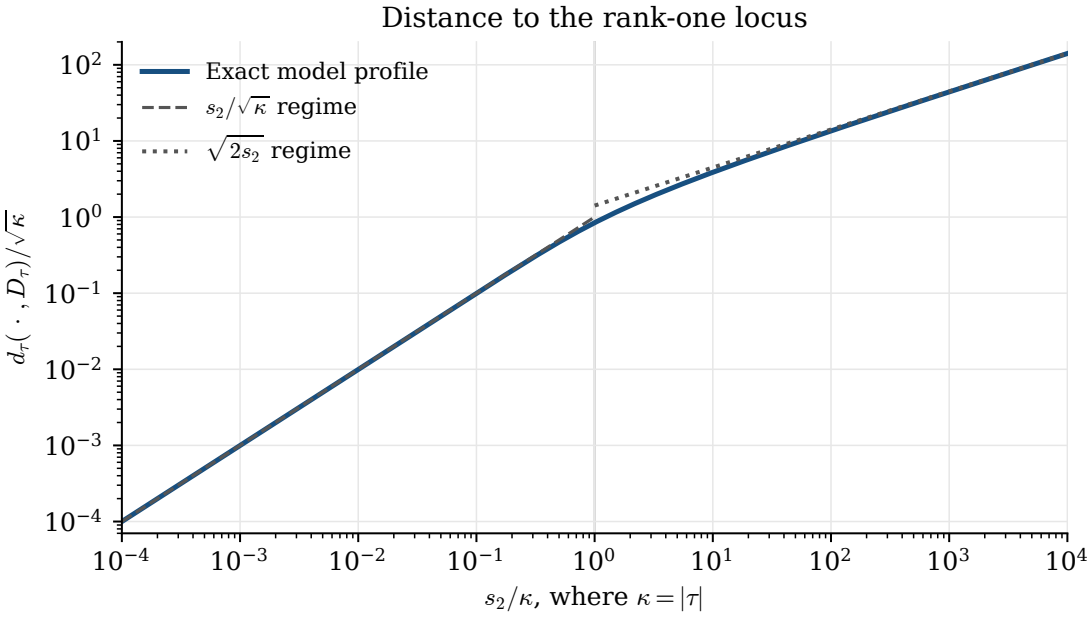


FIGURE 1. The exact model distance to D_τ, divided by $\sqrt{\kappa}$, as a function of s_2/κ, where $\kappa = |\tau|$. The two reference lines show the limiting regimes. The actual moduli distance has this profile with the relative error in Theorem 1.2. The plotted curve is evaluated from (7); it is not an empirical fit.

velocities are also compact-horizontal, although the ambient length upper bound already suffices.

For $\tau < 0$ use adjoint interchange. For $\tau = 0$, $a_i = b_i = \sqrt{s_i}$ and $F_0(b_i) = \sqrt{2}\, b_i$; the same paths reach lower strata in finite length. Inequality (21) holds on the pieces and passes to the limiting endpoints. Hence no path through a lower stratum shortens the displayed lower bounds. □

The scaling identity is $F_\kappa(\sqrt{\kappa}\, b) = \sqrt{\kappa}\, F_1(b)$. Consequently $F_\kappa(b_2) \sim s_2/\sqrt{\kappa}$ if $s_2/\kappa \to 0$, and $F_\kappa(b_2) \sim \sqrt{2s_2}$ if $s_2/\kappa \to \infty$. The full profile, rather than either asymptotic alone, is used below.

Lemma 3.3. *The radius r_τ is 1-Lipschitz. There is an absolute L such that any p, q of radius at most ρ can be joined inside $\{r_\tau \le \rho\}$ with length at most $L\rho$.*

Proof. The Euclidean radius is compact-invariant and 1-Lipschitz; subtracting a compact vertical vector leaves its differential unchanged. This proves the quotient assertion. For $\tau > 0$, write $A = U(\tau I + Q)^{1/2}$, $Q = BB^*$, and shrink B to uB, replacing A by $U(\tau I + u^2 Q)^{1/2}$. The squared speed is

$$\operatorname{tr} Q + \operatorname{tr}\big(u^2 Q^2 (\tau I + u^2 Q)^{-1}\big) \le 2 \operatorname{tr} Q.$$

The cost is at most r_τ and radius decreases. The two resulting points on the zero section can be joined at cost $D\sqrt{\tau}$, where D is the diameter of the fixed Grassmannian quotient metric. Since $r_\tau^2 \ge 2\tau$, $L = 2 + D/\sqrt{2}$ works. The negative case is symmetric. At zero join both points radially to the vertex, at total cost at most 2ρ. □

3.3. **Fibre metrics and comparison with the wall.** The zero-section metric is κ times its Stiefel quotient metric. For a rank-one matrix $N = suv^*$, with $s > 0$ and orthogonal unit u, v, put $W = (u, v)^\perp \simeq \mathbb{C}^3$. For $\tau > 0$ a point in its $\mathbb{P}(W)$ fibre has representative

$$A = (\sqrt{p}\,u, \sqrt{\tau}\,w), \qquad B = \begin{pmatrix} \sqrt{q}\,v^* \\ 0 \end{pmatrix}, \qquad \|w\| = 1, \quad p, q > 0, \quad p - q = \tau,\ pq = s^2.$$

A Hopf-horizontal variation $\dot{w} \perp w$ in W has sole variation $\dot{A}_2 = \sqrt{\tau}\,\dot{w}$. Every complex orbit variation has A-columns in $\langle u, w\rangle$, so this tangent is orthogonal to the whole complex vertical space. The induced fibre metric is exactly τg_{FS}, where g_{FS} is the unit-sphere Hopf quotient metric. This is independent of s. Adjoint interchange gives the negative case with factor κ.

Lemma 3.4. *On the common rank-two locus of M_τ and M_0, identified by N,*

$$e^{-\operatorname{arsinh}(|\tau|/(2s_2))} g_0^{\mathrm{q}} \le g_\tau^{\mathrm{q}} \le e^{\operatorname{arsinh}(|\tau|/(2s_2))} g_0^{\mathrm{q}}. \tag{22}$$

Proof. Choose a zero-level representative $A_0 = U\,\mathrm{diag}(\sqrt{s_1}, \sqrt{s_2})$, $B_0 = \mathrm{diag}(\sqrt{s_1}, \sqrt{s_2})V^*$. Its representative at level τ in the same complex orbit is $(A_0 e^S, e^{-S} B_0)$, with

$$S = \mathrm{diag}(\lambda_1, \lambda_2), \qquad \lambda_i = \tfrac{1}{2} \operatorname{arsinh}\big(\tau/(2s_i)\big). \tag{23}$$

This solves $2s_i \sinh(2\lambda_i) = \tau$ and preserves AB. Hermitian functional calculus defines S smoothly even at a repeated positive singular value.

On a central real moment level inside the complex equation, the quotient metric is also the infimum over the full *complex* orbit tangent. Indeed the imaginary compact-orbit directions are the gradients of the real moment components, hence are normal to the level. Compact vertical directions are orthogonal to them because the moment value is central. Their joint orthogonal complement is exactly the compact horizontal tangent.

Differentiating the point-dependent gauge in (23) adds a complex vertical vector to its linear action on a tangent. This is true without commutativity of S and its derivative: the logarithmic derivative of e^S is a Lie algebra element. The added vector vanishes in the complex vertical infimum. The linear action transports the vertical space and changes squared ambient norms by factors between $e^{-2\|S\|}$ and $e^{2\|S\|}$. Since $2\|S\| = \operatorname{arsinh}(|\tau|/(2s_2))$, this proves the result. □

The error in (22) is $O(|\tau|/s_2)$ when that ratio is small. Its loss at rank one is intentional: the complete inner model and the exact distance profile describe the region where this outer approximation is no longer small.

4. Harmonic deformation coordinates

Give L_i its ω_0-Hermitian–Einstein metric and E_0 the direct-sum metric h_0. The entire parallel trace-free algebra is the determinant-normalized copy of $\mathfrak{gl}_2$. All complements and Sobolev spaces below are fixed at this reference metric; the parameters vary on one sufficiently small common neighborhood.

4.1. **The harmonic pairing.**

Lemma 4.1. *The two cross-*Ext^1 *spaces have orthonormal Serre-dual bases for which the harmonic norm on* $\mathrm{Ext}^1(E_0, E_0)$ *is the Euclidean norm on $\mathcal{V}$ and the quadratic obstruction has zero locus $BA = 0$.*

Proof. Put $L = L_1^* \otimes L_2$, with its induced metric, and give L^* the dual metric. Normalize the parallel holomorphic two-form Ω so that in a unitary coframe

$$\omega_0 = i(\theta^1 \wedge \bar\theta^1 + \theta^2 \wedge \bar\theta^2), \qquad \Omega = \theta^1 \wedge \theta^2, \qquad \Omega \wedge \bar\Omega = dV_0.$$

For a local unitary frame e of L, define

$$J_L\big((a_1\bar\theta^1 + a_2\bar\theta^2)e\big) = (-\bar a_2\bar\theta^1 + \bar a_1\bar\theta^2)e^*.$$

This is frame-independent and antiunitary, with $J_{L^*}J_L = -I$. For the bilinear Serre pairing $Q_L(\alpha,\gamma) = \int_X \Omega \wedge \alpha \wedge \gamma$, the pointwise calculation gives

$$Q_L(\alpha, J_L\beta) = \langle \alpha, \beta\rangle_{L^2}, \tag{24}$$

where the Hermitian product is linear in the first argument.

If β is harmonic, then for every section s of L, $Q_L(\bar\partial_L s, J_L\beta) = 0$. Stokes implies $\bar\partial_{L^*}J_L\beta = 0$. For a section u of L^*,

$$\langle \bar\partial_{L^*}u, J_L\beta\rangle = -Q_{L^*}(\bar\partial_{L^*}u, \beta) = 0$$

by the same identity and Stokes. Thus $J_L\beta$ is coclosed as well. An orthonormal basis α_i of $H^{0,1}(L)$ and $\beta_i = J_L\alpha_i$ therefore have the required duality.

Write A with rows $u_i \in \mathrm{Hom}(\mathbb{C}^2,\mathbb{C})$ and B with columns $v_i \in \mathrm{Hom}(\mathbb{C},\mathbb{C}^2)$. The cross-term form

$$x = \begin{pmatrix} 0 & \sum_i \beta_i \otimes v_i \\ \sum_i \alpha_i \otimes u_i & 0 \end{pmatrix}$$

has norm $\|A\|^2 + \|B\|^2$. If η generates $H^{0,2}(\mathcal{O}_X)$ with $\int_X \Omega \wedge \eta = 1$, the class of $x \wedge x$ has blocks $-BA\,\eta$ and $\mathrm{tr}(BA)\eta$. Using $\eta = \bar\Omega$ instead multiplies both blocks by $1/\operatorname{Vol}(X,\omega_0)$. The zero equation is unchanged. □

The parallel normalization is where the Ricci-flat central metric is used. A general Kähler metric need not make the holomorphic two-form have constant norm. The compatibility with the hyperkähler harmonic pairing is also described in [Ver96, arXiv version, Proposition 9.1].

4.2. **Convergent equivariant quadratic coordinates.** Let $d = \bar\partial_0$ on trace-free endomorphism-valued forms, and let $\mathcal{H}^1$ be its harmonic degree-one space. The standard harmonic Kuranishi embedding Φ satisfies

$$\Phi(b) = b - d^*G_2(\Phi(b)\wedge\Phi(b)), \qquad D\Phi(0) = \iota, \tag{25}$$

where G_2 is the Green operator in degree two and ι the harmonic inclusion. It is holomorphic and equivariant for parallel automorphisms. Its integrability equation is the harmonic projection of $\Phi(b)\wedge\Phi(b)$; see [BS22, Theorem 1] and [Del25, Proposition 4.3]. We use reduced analytic spaces.

Proposition 4.2. *There is a G-equivariant analytic isomorphism from the reduced germ $\mathcal{Q} = \{BA = 0\}$ to the full reduced harmonic integrability germ. It has an equivariant ambient holomorphic extension F satisfying*

$$F(x) = x + O(|x|^3), \qquad DF(x) = I + O(|x|^2). \tag{26}$$

Proof. Formality follows from [BZ19, Theorem 1.2], because E_0 is H-Gieseker-polystable. The equivariant analytic result [AS18, Proposition 4.4], with its quiver identification, gives an isomorphism $\psi : Z \to \mathcal{Q}$ from a Quot slice, tangent to the identity. We must compare this slice with the *harmonic* family; formality alone does not supply that comparison.

Write S for the reduced harmonic base. The family on $X \times S$ is holomorphic and analytically semiuniversal in the reduced category [BS21, arXiv version, §5 and Remark 4]. Retain

the full automorphism group $\widetilde{G} = GL_2 \times GL_1$ while acting on this family. For m large, the twist of the central bundle by mH is generated by sections and has no higher cohomology. After shrinking S, proper base change makes the pushforward of the family a holomorphic vector bundle with fibre $W = H^0(X, E_0(mH))$, and evaluation remains surjective. Choose a frame T_s equal to the identity at zero. Average over a maximal compact $\widetilde{K}$:

$$T_s^{\widetilde{K}} = \int_{\widetilde{K}} \rho(k)^{-1} T_{ks}\, k_{\mathcal{W}}\, dk.$$

This is a holomorphic equivariant frame after shrinking. The evaluation quotient defines an analytic equivariant map $q : S \to R$ to the open algebraic Quot locus, by the analytic Quot theorem [Gro61, Théorème 1.1]. Scalars act ineffectively on the base. Holomorphic compact equivariance complexifies as an identity of germs.

Let $\Gamma = GL(W)$. Choose an $\widetilde{G}$-invariant complement $\mathfrak{m}$ to its stabilizer Lie algebra in $\operatorname{Lie}\Gamma$. The Luna map $\Gamma \times^{\widetilde{G}} Z \to R$ is étale [Lun73, §III.1]. The exponential local section of $\Gamma/\widetilde{G}$ gives an analytic isomorphism of germs $(u, z) \mapsto e^u z$ from $\mathfrak{m} \times Z$ to R. This remains true for singular reduced germs. Projecting its inverse after q yields an equivariant analytic map $h : S \to Z$. The factor $e^{u(s)}$ identifies the two marked families, so the Kodaira–Spencer maps give $Dh(0) = I$.

Both reduced tangent spaces are all of $\mathcal{V}$. The pure-arrow subspaces $(A, 0)$ and $(0, B)$ lie in $\mathcal{Q}$ and span $\mathcal{V}$. They also lie in S: their harmonic forms have square zero, and the unique solution of (25) is the form itself. Reduced semiuniversality now gives a reverse analytic classifying map $g : Z \to S$ with $Dg(0) = I$, without requiring g to be equivariant.

An analytic germ self-map with invertible tangent map is an automorphism. Indeed, in a minimal ambient embedding its lift H is a local biholomorphism. The induced endomorphism of the local ring is surjective, since an ambient representative f has preimage $f \circ H^{-1}$. A surjective endomorphism φ of a Noetherian ring is injective: the chain $\ker \varphi^n$ stabilizes, and lifting an element of $\ker \varphi$ by a sufficiently high surjective power then shows it is zero. Applying this to gh and hg proves that h is an isomorphism. Thus $F = h^{-1}\psi^{-1}$ is an equivariant analytic comparison. No nonreduced formal miniversal hull, or convergence of an arbitrary formal series, is used.

Extend the coordinate functions of F to an ambient map F_0. Its derivative is I because the reduced tangent space is all of $\mathcal{V}$. Averaging $\int_K k^{-1} F_0(kx)\, dk$ preserves its restriction to $\mathcal{Q}$ and makes it equivariant. The ambient inverse-function theorem applies. The compact automorphism $\operatorname{diag}(-I_2, 1)$ acts as $-I$ on $\mathcal{V}$, so the averaged map is odd. This proves (26). □

Set

$$a(x) = \Phi(F(x)), \qquad D_x = d + [a(x), \cdot\,], \qquad J_x = Da(x). \tag{27}$$

These maps are defined on a full ambient ball. If $C_x\xi$ is the infinitesimal coefficient action of a parallel complex endomorphism, equivariance gives the exact identity

$$D_x\xi = -J_x C_x \xi. \tag{28}$$

For a tangent vector to a smooth piece of $\mathcal{Q}$, $D_x J_x v = 0$.

4.3. **The corrected real equation.** Let $dV_0 = \omega_0^2/2$ and use its fixed pairing on zero-forms. For the h_0-unitary connection associated to a Dolbeault operator ∂, define the Hermitian-valued moment

$$M_t(\partial) = -\frac{i(F_\partial^0 \wedge \omega_t)}{dV_0}. \tag{29}$$

The superscript denotes trace-free curvature. Its zero equation is projective HYM; imposing the determinant line's HE metric separately gives the fixed-determinant HYM equation. With the affine symplectic form associated to (2), integration by parts gives

$$d\langle M_t, h\rangle(v) = 2\Omega_{\infty,t}(\xi_h, v), \qquad \xi_h = -D_{\partial}(ih). \tag{30}$$

For example, the quadratic $(1,1)$ curvature is $-a \wedge a^\dagger - a^\dagger \wedge a$; the $-iF$ moment has multiplicity-two block $A^*A - BB^*$.

Proposition 4.3. *There is a unique small smooth Hermitian correction $\sigma(t,x)$, orthogonal to the full parallel trace-free Hermitian algebra, such that*

$$\Psi_t(x) = e^{\sigma(t,x)} \cdot (\bar\partial_0 + a(x)), \qquad M_t(\Psi_t(x))$$

has only its parallel component. The correction is K-equivariant and, in every fixed finite collection of spatial norms,

$$\|\sigma(t,x)\|_{C^k} \le C_k(|t| + |x|^2). \tag{31}$$

The pulled-back form $\Omega_t = \Psi_t^*\Omega_{\infty,t}$ and centered finite moment ρ_t satisfy

$$\|\Omega_{t,x} - \omega^{\mathrm{amb}}\| \le C(|t| + |x|^2), \qquad d\,\mathrm{tr}(\rho_t h)(v) = 2\Omega_t(\xi_h, v), \qquad \rho_t(0) = 0. \tag{32}$$

In these coordinates the actual HYM equation is precisely

$$\rho_t(x) = \tau(t) I_2, \qquad \tau(t) = -28\pi t/3. \tag{33}$$

Proof. Project (29) onto the fixed orthogonal complement of the parallel Hermitian algebra, after applying a positive gauge e^s. At $(t,x,s) = (0,0,0)$ its derivative in s is a nonzero fixed multiple of the covariant Laplacian on that complement. The entire kernel has been removed, so the derivative is an elliptic isomorphism between fixed high-regularity Sobolev spaces. The implicit-function theorem and elliptic regularity give the smooth correction and uniqueness. Its x derivative at zero vanishes by the Kähler identities and harmonicity of $J_0 v$. This proves (31); uniqueness gives compact equivariance. This is also the HYM specialization of the construction in [Del25, Proposition 4.5].

The pullback of the affine symplectic form is closed and has the finite moment identity. At $(0,0)$ its value is ω^{amb}, since $D\Psi_0(0)$ is harmonic inclusion. Invariance under the automorphism acting as $-I$ makes its coefficient functions even in x. Smoothness gives the first estimate in (32). It is symplectic and tames I on a smaller ball. It need not be compatible with I.

For $h \in \mathrm{Herm}_2$, the determinant-one lift is

$$\widehat{h} = \mathrm{diag}(S, c), \qquad S = h - \tfrac{1}{3}\,\mathrm{tr}(h) I_2, \quad c = -\tfrac{1}{3}\,\mathrm{tr}(h). \tag{34}$$

Define m_t by $\mathrm{tr}(m_t(x)h) = \langle M_t(\Psi_t(x)), \widehat{h}\rangle_{L^2(dV_0)}$. At $x = 0$ the correction is block scalar by compact invariance, so the integrated pairing with each parallel block is topological. With $c_1(L) = i[F_L]/(2\pi)$ and (16),

$$\begin{aligned}\mathrm{tr}(m_t(0)h) &= -2\pi\big(\deg_t L_1 \ \mathrm{tr}\, S + \deg_t L_2\, c\big) \\ &= -\tfrac{2\pi}{3}\big((3-7t) - (3+7t)\big)\,\mathrm{tr}\, h = \tfrac{28\pi}{3} t\ \mathrm{tr}\, h.\end{aligned}$$

The fixed density cancels in this integral; no volume factor is present. Set $\rho_t = m_t - m_t(0)$. Its moment identity has the normalization in (32), and vanishing of the actual moment is exactly (33). □

For $t < 0$, the positive real level is the A-injective chamber. This sign also agrees with the slope test for an A-only extension: its subbundle L_2 has lower slope than E precisely when $t < 0$.

5. Uniform correction of the real moment map

This section proves the estimate that permits independent degeneration scales. It applies to the full complex equation $\mathcal{Q}$, including its lower-rank points. Compatibility of the corrected form with the coefficient complex structure is not assumed.

Theorem 5.1. *Let Ω_t be a smooth family of closed K-invariant forms on a ball in $\mathcal{V}$, with centered moment maps ρ_t in the convention of (32), and suppose*

$$\|\Omega_{t,x} - \omega^{\mathrm{amb}}\| \le C(|t| + |x|^2). \tag{35}$$

For every sufficiently small $x = (A, B) \in \mathcal{Q}$ with $\mu(x) = \tau I$, there is a Hermitian matrix S such that

$$T_S x = (Ae^S, e^{-S}B) = y, \qquad \rho_t(y) = \tau I, \qquad \|S\| \le C'(|t| + |x|^2). \tag{36}$$

At $\tau = 0$, S may be taken zero on the common kernel of A and B^*, with no off-diagonal block to its orthogonal complement.

The map on compact orbits defined by the common complex orbit is unique, continuous through rank change, and smooth on each fixed-rank piece. It covers all sufficiently small points of the corrected real level, after a fixed adjustment of the domain radius. No constant depends on the smaller positive eigenvalue of $A^*A + BB^*$.

Proof. Write $E = \rho_t - \mu$. On a ball where $\|\Omega_t - \omega^{\mathrm{amb}}\| \le \epsilon$, every orthogonal projection P on $\mathbb{C}^2$ satisfies

$$|\operatorname{tr}(PE(x))| \le \epsilon\big(\|AP\|^2 + \|PB\|^2\big). \tag{37}$$

To see this, decompose $x = n + u$ into the fixed and moving parts for the circle generated by P:

$$n = (A(1-P), (1-P)B), \qquad u = (AP, PB).$$

Each moment component is constant on the connected fixed linear subspace and has value zero there. On $n + su$, $\xi_P(n + su) = s\xi_P(u)$ and $\|\xi_P(u)\| = \|u\|$. Hence

$$\operatorname{tr}(PE(x)) = 2\int_0^1 s(\Omega_{t,n+su} - \omega^{\mathrm{amb}})(\xi_P(u), u)\, ds,$$

which proves (37). The segment stays in the same ball. Testing rank-one projections also gives the matrix inequality $-\epsilon M \le E \le \epsilon M$, where $M = A^*A + BB^*$. This relative error is stronger than an additive bound near rank loss.

Interpolate $\Omega_{t,s} = (1-s)\omega^{\mathrm{amb}} + s\Omega_t$ and $\rho_{t,s} = \mu + sE$ for $0 \le s \le 1$. They tame I on a small common ball. Suppose first $\tau \neq 0$, so the action at x is free. Starting from $S = 0$ at $s = 0$, solve $\rho_{t,s}(T_S x) = \tau I$. The derivative in imaginary orbit directions is invertible, because for every nonzero Hermitian h,

$$d\operatorname{tr}(\rho_{t,s}h)(I\xi_h) = 2\Omega_{t,s}(\xi_h, I\xi_h) > 0.$$

This positive quadratic pairing gives injectivity; equal dimensions give invertibility, without requiring symmetry of the derivative. Positive polar coordinates give the same invertibility. More explicitly, the Hermitian part of the logarithmic derivative of e^S has, in an eigenbasis of S, coefficients $\sinh(\lambda_i - \lambda_j)/(\lambda_i - \lambda_j)$, interpreted as one on the diagonal. They are positive.

The remaining compact part has zero moment derivative at a central value. This argument asserts invertibility at each free point, not a uniform inverse bound.

The uniform bound instead follows from (37). Let P be the spectral projection onto a unit eigenvector of S, with eigenvalue λ, and put $a = \|AP\|^2$, $b = \|PB\|^2$ at the original balanced point. Thus $a - b = \tau$. At a solution,

$$0 = (e^{2\lambda} - 1)a + (1 - e^{-2\lambda})b + s\,\mathrm{tr}(PE(T_S x)). \tag{38}$$

The last term is at most $\epsilon(e^{2\lambda}a + e^{-2\lambda}b)$ in absolute value. For $0 < \epsilon < 1/10$, both lower-bound coefficients are positive when $\lambda > 2\epsilon$, and both upper-bound coefficients are negative when $\lambda < -2\epsilon$. Since $a + b > 0$, neither is possible. Thus $\|S\| \le 2\epsilon$.

Use the ball of radius $2|x|$, with $\epsilon = C(|t|+4|x|^2)$. For small t, x, the bound ensures $|T_S x| \le e^{2\epsilon}|x| < 2|x|$. A continuation branch cannot leave the domain or escape in S. Compactness gives a solution at each limiting interpolation time, and the invertible linearization continues it. This proves (36).

If $\tau = 0$, flat balance gives $A^*A = BB^* = P_0$. Restrict to its support V_0. Both maps vanish on the complementary kernel, and the $U(V_0)$ action on the support is free. Apply the same argument there. Equivariance under $U(V_0^\perp)$ makes the moment block diagonal; its kernel block is zero by (37), and the off-diagonal block vanishes by invariance. The restricted solution therefore solves the full equation. At $x = 0$, take $S = 0$.

To prove uniqueness of the compact orbit, take two small corrected balanced points in the same complex orbit. After a polar decomposition and compact change they are $z, T_H z$ for Hermitian H. The squared norm along $T_{uH}z$ is a sum of positive exponentials, hence is convex and bounded by the larger endpoint norm. The path stays in the common ball. Pairing the difference of the two central moment values with H gives

$$0 = 2\int_0^1 \Omega_t(\xi_H, I\xi_H)(T_{uH}z)\,du.$$

Taming forces $\xi_H = 0$, so the positive transformation fixes the point. This also gives uniqueness of the positive representative on the free support.

For coverage, let a small $y = (A, B) \in \mathcal{Q}$ satisfy $\rho_t(y) = \tau I$. If $\tau > 0$, testing (37) on a vector in $\ker A$ is impossible, so A is injective. For $\tau < 0$, B is surjective. At zero, the same argument gives $\ker A = \ker B^*$. In each case the orbit has a flat-balanced representative by Proposition 3.1. For nonzero level the construction preserves the incidence pair $(\mathrm{im}\,A, AB)$ or $(\ker B, AB)$, respectively. At zero, restrict to the common support: two full-support factorizations of the same AB differ by a general linear transformation on that support. It stays small: taking traces and using the nuclear norm,

$$2|\tau| \le (1+\epsilon)|y|^2, \qquad 2\|AB\|_* \le |y|^2.$$

Its flat squared radius is $\sum_i \sqrt{\tau^2 + 4s_i^2} \le (2+\epsilon)|y|^2$. A fixed shrinking of the actual domain thus gives coverage.

On each fixed-rank piece, choose a smooth support frame and use the implicit-function theorem. For continuity at rank loss, take a convergent sequence of flat representatives. The matrices S stay bounded, and each subsequential limit gives a corrected representative in the limiting complex orbit. Compact-orbit uniqueness identifies all such limits. For the inverse, the preceding radius bound makes the flat representatives bounded; pass to a subsequence and use the same bounded gauges and compact unitary identifications. Its only possible limiting orbit is the flat representative of the given corrected limit. This proves inverse continuity without assuming continuity of the selected matrices S. □

Corollary 5.2. *Suppose the actual quotient norm on the corrected locus is*

$$Q_{t,y}([w]) = \inf_{\eta\in\mathfrak{gl}_2(\mathbb{C})} \mathcal{B}_{t,y}(w - C_y\eta, w - C_y\eta),$$

where $\mathcal{B}$ is a positive Hermitian form on the ambient coefficient space and

$$(1 - C(|t| + |y|^2))g^{\mathrm{amb}} \le \mathcal{B}_{t,y} \le (1 + C(|t| + |y|^2))g^{\mathrm{amb}}. \tag{39}$$

Then the common-complex-orbit map compares Q_t with g^{q}_τ with relative error $O(|t| + |x|^2)$ on every smooth quotient piece, uniformly as the positive eigenvalues of $A^*A + BB^*$ tend to zero.

Proof. Differentiate $y = T_{S(x)}x$ on a smooth piece. Its derivative is $T_S v$ plus a vector in the full complex vertical tangent. That vector disappears in the infimum, without any estimate on dS. The linear action T_S carries one complex vertical space onto the other and distorts squared ambient norms by $e^{\pm 2\|S\|}$. Apply (39) before taking the infimum, and use Theorem 5.1. At the flat level the complex-vertical infimum is the quotient metric, as in Lemma 3.4. □

The additional norm hypothesis in this corollary is essential. A tame symplectic form alone does not identify the harmonic metric. We verify that hypothesis for the actual HYM family next.

6. The actual harmonic metric

We now identify the form required in Corollary 5.2 with harmonic cohomology. The uniformly invertible operator in this argument is the infinite-dimensional complementary gauge operator. The possibly small finite orbit directions are retained until the last minimization.

6.1. **The complementary projection.** On the uncorrected holomorphic family $\bar\partial_0 + a(x)$, put

$$H_{t,x} = h_0 e^{2\sigma(t,x)}. \tag{40}$$

The gauge $e^{\sigma(t,x)}$ is an isometry from this metric to h_0 on the corrected presentation. At an integrable zero of the corrected moment, $H_{t,x}$ is the actual projectively HE metric; the scalar determinant adjustment has no effect on endomorphism norms. Away from that locus it is a smooth positive auxiliary metric.

Let $\mathfrak{g}$ be the full parallel complex trace-free algebra, and let Q be its fixed L^2 complement in trace-free zero-forms. We write $QW^{1,2} = Q \cap W^{1,2}$. At the central bundle, ellipticity and removal of the kernel give

$$\|\eta\|_{W^{1,2}} \le C\|d\eta\|_{L^2}, \qquad \eta \in QW^{1,2}. \tag{41}$$

Since $a(x)$ is small in C^0, this persists with D_x in place of d. The norms for $H_{t,x}, \omega_t$ remain uniformly equivalent.

Proposition 6.1. *There is a smooth positive K-equivariant Hermitian form $\mathcal{B}_{t,x}$ on the whole ambient coefficient space satisfying*

$$(1 - C(|t| + |x|^2))g^{\mathrm{amb}} \le \mathcal{B}_{t,x} \le (1 + C(|t| + |x|^2))g^{\mathrm{amb}}, \tag{42}$$

such that

$$\inf_\zeta \|J_x v - D_x\zeta\|^2_{H_{t,x},\omega_t} = \inf_{\xi\in\mathfrak{g}} \mathcal{B}_{t,x}(v - C_x\xi, v - C_x\xi). \tag{43}$$

On an integrable HE point and a tangent to a smooth reduced quotient piece, the left side is its actual harmonic Kodaira–Spencer norm.

Proof. Define the form by polarization of

$$\mathcal{B}_{t,x}(v,v) = \inf_{\eta \in QW^{1,2}} \|J_x v - D_x \eta\|^2_{H_{t,x},\omega_t}. \tag{44}$$

The range $D_x QW^{1,2}$ is closed by (41). More explicitly, on the fixed Hilbert space $\mathcal{H} = QW^{1,2}$ the normal operator is

$$L_{t,x} : \mathcal{H} \to \mathcal{H}^*, \qquad \langle L_{t,x}\eta, \nu\rangle = \langle D_x\eta, D_x\nu\rangle_{H_{t,x},\omega_t}.$$

It is uniformly bounded and coercive. Its inverse varies smoothly by the bounded-inverse identity. The minimizing η_v solves $L_{t,x}\eta_v(\nu) = \langle J_x v, D_x\nu\rangle$. Thus the orthogonal projection, and hence $\mathcal{B}$, are smooth on the full ambient ball, without a rank condition on an obstruction map or finite orbit map.

At $(0,0)$, harmonic orthogonality gives $\mathcal{B}_{0,0} = g^{\mathrm{amb}}$, so it is positive nearby. All ingredients are compact-equivariant, including the fixed complement. The automorphism acting by $-I$ on $\mathcal{V}$ gives $\mathcal{B}_{t,-x}(-v,-w) = \mathcal{B}_{t,x}(v,w)$. Its coefficient matrix is therefore even in x. Taylor's theorem gives (42). This parity argument is a simplification available in the present representation.

Split every complex zero-form gauge as $\zeta = \xi + \eta$ with $\xi \in \mathfrak{g}$, $\eta \in Q$. Equation (28) permits the first minimization over η to be performed exactly, leaving $\mathcal{B}(v + C_x\xi, v + C_x\xi)$. Replace ξ by $-\xi$ to obtain (43). No inverse of the finite orbit operator has been used.

At an integrable point and for a tangent along its reduced integrability piece, $D_x J_x v = 0$. Hodge theory identifies the least norm in this Dolbeault cohomology class with its harmonic norm. Changing to the corrected presentation by a varying gauge g changes the representative to

$$g(J_x v)g^{-1} - D'_x(\dot{g}g^{-1}).$$

Conjugation is an isometry for the transported metric, and the second term is exact. Thus derivatives of σ do not enter the norm identity. The interpretation concerns genuine tangent classes, not arbitrary Zariski tangent vectors at a singular coarse quotient. □

Together, Proposition 4.3, Theorem 5.1 and Proposition 6.1 meet every hypothesis of Corollary 5.2. They prove the tensor estimate (5) on the corrected quotient, with constants independent of the positive eigenvalues of $A^*A + BB^*$.

6.2. **The full actual moduli neighborhood.**

Proposition 6.2. *The corrected integrable solution quotient describes an open neighborhood in actual fixed-determinant HYM moduli. Combined with Theorem 5.1, it gives the maps f_t of Theorem 1.1, on a common radius domain. The nonzero chambers consist of stable bundles, and the rank-one wall piece consists of $L_1 \oplus F$ with F stable of vector $(2, H, 0)$.*

Proof. The harmonic slice is a complete local deformation family, with local complex-orbit classification [Del25, Proposition 4.3]. On the integrable locus, the corrected finite solution quotient has an open image in actual HYM moduli [Del25, Lemma 7.3 and Theorem 7.4]. We spell out why our fixed-density and complementary normalization describes that same full neighborhood.

Put any sufficiently close integrable solution into the harmonic family by complex Coulomb gauge. The small gauge taking a harmonic representative to the actual solution has polar decomposition ke^s. The unitary part does not change the moduli point. Move the harmonic base point by a small positive parallel gauge e^h and polar-decompose $e^s e^{-h}$. The projection onto the parallel Hermitian algebra of the logarithm of its positive factor has derivative $-I$ in h at $(s,h) = (0,0)$. The finite implicit-function theorem therefore makes that logarithm

complementary. Uniqueness of the complementary equation in Proposition 4.3 identifies it with $\sigma(t,x)$. All operations are continuous on common parameter and coefficient neighborhoods. Conversely, every integrable zero of this corrected slice is an actual solution. Thus the normalization changes the slice, not its solution quotient.

Local complex-orbit classification and the compact-orbit uniqueness in Theorem 5.1 give a well-defined continuous inverse. The inverse coverage in that theorem permits a fixed shrinking of the radius. Work inside a larger common chart; the image of any smaller strict radius domain is open. The finite-regularity slice topology agrees here with the smooth connection quotient topology: the finite representatives depend continuously in every fixed spatial norm by elliptic bootstrapping, and their inverse is already continuous in a fixed sufficiently high regularity. No weak compactification topology is used.

We also identify stabilizers. If a small reconstructed bundle has a holomorphic trace-free endomorphism $\xi+\eta$, with $\xi\in\mathfrak{g}$ and $\eta\in Q$, then $D_x\eta = J_xC_x\xi$. Equation (44) gives $\mathcal{B}(C_x\xi, C_x\xi)=0$. Positivity forces $C_x\xi=0$, and complementary coercivity gives $\eta=0$. Thus the actual infinitesimal automorphisms are exactly the finite stabilizer directions. At nonzero level the effective finite action is free; the bundle is simple and polystable, hence stable by the Hermitian–Einstein correspondence [Don85, UY86].

The complete holomorphic deformation family induces the usual holomorphic classifying map to stable-bundle moduli. Its Kodaira–Spencer map after quotienting is an isomorphism, so the local bijection is biholomorphic. Restriction to a fixed stabilizer piece gives the same conclusion there at the wall. The reduced analytic wall quotient is the invariant quotient of the exact quadratic germ, with coordinate $N=AB$, as established in Propositions 3.1 and 4.2.

At a rank-one point of the zero-level quotient, the unused multiplicity line defines a holomorphic idempotent. Equivariance preserves the splitting in the harmonic reconstruction and in the complementary correction. The corresponding line is L_1, since it has no self-deformations. The other summand has vector $(2,H,0)$ and no trace-free automorphisms on its effective support. It is therefore stable. Its dimension is $H^2+2=8$, in agreement with the rank-one stratum. At the vertex the effective stabilizer is $U(2)$.

Finally, the zero section at nonzero parameter has radius $\sqrt{2|\tau|}$, so its whole compact fibre lies in the common chart for small t. The incidence maps now give the stated local stratified Mukai flop. Section 2 identifies the nearby integral-polarization chambers with the locally free open loci in ordinary Gieseker moduli. No global wall-Gieseker morphism has entered the argument. □

This proves the topological and tensor assertions of Theorem 1.1. The remaining issue is intrinsic distance: a comparison on a coordinate domain alone does not prevent an ambient path from leaving it and returning at smaller cost.

7. Intrinsic geometry of the contraction

7.1. **Excluding paths outside the chart.** Choose the radius R inside a larger common chart and small enough that (5) compares lengths by factors between $1/2$ and 2. The closed model set $\{r_\tau \le R\}$ is compact: it is the compact quotient of a closed bounded finite-dimensional moment level. Its image in the Hausdorff unitary-gauge moduli space is compact and closed. A continuous path leaving the open radius-R image must therefore first meet its radius-R boundary.

Until this first exit, Lemma 3.3 and the tensor estimate give

$$|r_\tau(q) - r_\tau(p)| \le 2 \operatorname{length}_{g_t}(\gamma). \tag{45}$$

These inequalities extend through finite-length lower-stratum endpoints by continuity.

Proof of (6). Let both endpoints have radius at most ρ. Their model connecting path from Lemma 3.3 has length at most $L\rho$ and stays inside that radius. It gives an actual competitor of length at most $2L\rho$. An escaping path pays at least $(R-\rho)/2$ by (45). For $\rho \le cR$ with c sufficiently small, such a path cannot improve the infimum. Every remaining actual near-minimizer can be taken of length at most $4L\rho$ and hence stays at radius at most $(1+8L)\rho$.

Conversely, a global model near-minimizer can be chosen with length at most $2L\rho$. The model radius inequality keeps it within $(1+2L)\rho$. On these enlarged but small domains the sharp relative tensor error is $O(|t|+\rho^2)$. Apply it to both sets of paths and take infima, obtaining (6). No minimizer need exist: use paths within an arbitrary positive tolerance and pass to zero tolerance.

For arbitrarily close endpoints, the coarse radius bound first confines all candidates; the sharp multiplicative comparison is then applied to their entire lengths. Thus there is no additive $O(\rho)$ error and no division by their possibly tiny distance. The model distance separates points, so the actual distance also separates points in this neighborhood. □

7.2. **Distances to the exceptional loci.**

Proof of Theorem 1.2. The model paths of Proposition 3.2 decrease radius and have length at most the starting radius r. Applying (5) gives the upper bounds in (8)–(9); their endpoints lie in the stated local target sets.

Any actual competitor capable of lowering these bounds can be taken of length at most $2r$. For $r \le cR$ it cannot leave the chart, by the same exit-cost argument, and (45) confines it to radius $5r$. Its inverse image ends in the appropriate model rank locus, so its model length is bounded below by the *global* distance in (20). The sharp tensor comparison on radius $5r$ proves the lower bounds. This remains multiplicative when the smaller distance is arbitrarily small compared with r.

On the zero section, $r_\tau^2 = 2|\tau|$, so the induced metric comparison has relative error $O(|t|)$. On the rank-one fibre over singular value s, equation (18) gives $r_\tau^2 = \sqrt{\tau^2+4s^2} + |\tau| = O(|t|+s)$. The exact induced fibre metrics computed in Section 3 therefore give the asserted Grassmannian and projective-plane formulas. □

The collapse length scale is consequently $\sqrt{|\tau|} = \sqrt{28\pi|t|/3}$. Both the deepest Grassmannian and the $\mathbb{P}^2$ fibres over rank one have this scale, even while the rank-one point approaches the vertex at an unrelated rate. The error in the actual metric remains controlled by the total deformation radius, without requiring a positive lower bound for the smaller transverse distance.

7.3. **The complete inner metric and outer overlap.**

Proof of Corollary 1.3. Put $\kappa = |\tau|$ and write $x = \sqrt{\kappa}\,\widehat{x}$. The model homothety gives $g_\tau^{\mathrm{q}} = \kappa g_{\pm 1}^{\mathrm{q}}$ and $r_\tau^2 = \kappa \widehat{r}^2$. Since $|t| = 3\kappa/(28\pi)$, substitution into (5) proves (10). Each bounded-radius region of the complete model is eventually included; the same holds on any expanding radius R_t with $\kappa R_t^2 \to 0$.

For the pointed metric statement, fix a zero-section base point. The physical cost of leaving the radius-R chart stays bounded below independently of t. After division of lengths by $\sqrt{\kappa}$, that cost diverges. Equation (45) also bounds $\widehat{r}$ on every bounded rescaled actual ball about

the base point. Thus all such balls lie in controlled compact subcharts. On them (6) has relative error $O(\kappa)$, and every fixed bounded model ball is covered. These estimates give pointed Gromov–Hausdorff convergence of bounded balls. If balls of exactly equal radius are used, the $1 + O(\kappa)$ distance distortion changes radius only by $O(\kappa)$; move along a model minimizing geodesic to obtain the usual correspondence. Properness and completeness of the model were proved in Proposition 3.1. No completeness of the full actual bundle-moduli component is required.

On the common rank-two complex locus, Lemma 3.4 gives relative error $O(|\tau|/s_2)$. If that ratio is bounded and small, then (18) gives $r_\tau^2 = O(r_0^2)$. Composition with (5) proves (11) first relative to g_0^{q}. Applying (5) also at $t = 0$ compares the actual chamber and actual wall metrics in their common N coordinates. The overlap region $|\tau| \ll s_2 \le s_1 \ll 1$ matches the complete inner model to the wall. At s_2 comparable to or smaller than $|\tau|$, the inner estimate and full distance profile apply instead. □

All convergence and matching assertions above are statements about quadratic forms, lengths and metric spaces. Differentiated estimates would be needed for curvature or smooth convergence. The argument also uses a fixed smooth bundle near its reducible splitting; it does not include a non-locally-free or bubbling boundary. Within this neighborhood, however, the entire contraction and its change of stabilizer are described by the actual harmonic metric with the same relative error.

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